

Mathematical Frontiers in the OpenAI Research Release: Cancellation, Positivity, and Changes of Representation

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Abstract

The OpenAI mathematics release includes manuscripts claiming solutions to longstanding problems. Their significance depends both on the barriers their conclusions would cross and on the mathematics introduced to cross them. This comparative narrative review examines eight cases, spanning arithmetic, convex and operator inequalities, counting and sampling, computational hardness, group algebras, and kinetic equations. Each case separates the claimed change in the theorem from the step intended to achieve it and compares that step with its closest primary-literature antecedents. The most interesting proposed advances concern cancellation at every averaging cutoff, interpolation weights chosen independently of approximation centres, compatible estimates under matrix amplification, and signed force bounds strong enough for global continuation. Several surrounding tools, including similarity theory, residual sampling correction, and matrix-to-scalar transfer, are already established. Worked examples show why these distinctions matter: logarithmic cancellation can miss an ordinary-average bias, restricted energy control can coexist with slow mixing, and finite positive certificates need control between their nodes. Across the cases, cancellation, positivity, and changes of representation explain where the manuscripts seek additional power. The resulting comparison identifies candidate methodological advances and the precise mathematical questions on which their broader significance depends.

Keywords: mathematical review; cancellation; positivity; Diophantine approximation; approximate counting; proof verification.

1 Introduction

An unexpected solution is most informative when it reveals why an earlier obstruction disappears. A stronger theorem may result from a new estimate, an altered state space, a sharper quantifier, or a construction in which two familiar objects acquire a previously unavailable compatibility. These possibilities call for different assessments of mathematical novelty. They also determine which parts of a proof may be reusable elsewhere. Here a *candidate breakthrough* crosses a previously unresolved qualitative barrier, reaches a conjectured sharp endpoint, or changes the complexity status of a broad problem class. This distinguishes it from an improvement within an existing regime, such as a better nonsharp constant or a faster algorithm for an already tractable class. Proof novelty is a separate question: established tools can yield a major theorem once a difficult new estimate or compatibility is established.

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The OpenAI collection contains 722 manuscripts in 372 result families, distributed across 17 mathematical subject groups [68]. The repository describes a mixture of verification stages and supplies manuscripts, supporting formal artifacts, and selected reasoning summaries. This breadth creates a useful occasion for a comparative review, provided each case is considered through its precise statement and proof mechanism. Throughout this paper, a *release claim* denotes a conclusion stated in an accompanying manuscript; established results from the primary literature supply the comparison baseline. Mathematical consequences that depend on a release claim retain that dependence.

The selected cases address recognizable obstacles and give enough detail to locate the mathematical step carrying the claimed advance. They are connected by how their proofs organize difficult information: retain a signed sum, select the right notion of positivity, or move to a representation where a missing estimate becomes accessible. Table 1 provides the overview; the following sections explain the comparisons and work through the distinctions that theorem titles leave hidden.

2 Overview of the selected mathematical frontiers

The claimed conclusions in Table 1 would cross major qualitative boundaries if established. The last two columns separate methods already present in the literature from the additional estimate or construction proposed in the release. The comparisons are developed in the case studies below; the table is a guide to those arguments.

Table 1: Overview of the selected claims and the mathematical steps carrying their proposed advances.

Case	Barrier the conclusion would cross	Inherited or preceding machinery	Proposed step carrying the advance
Multiplicative correlations (007)	Cancellation at every ordinary cutoff for fixed forms.	Divisibility graphs, composite prime steps, circuit fooling, and earlier exceptional-scale estimates.	Shared-residue finite-law comparison and signed high-trace control uniform in the growing prime-supply count.
Irrationality exponent of π (017)	The optimal endpoint $\mu(\pi) = 2$, beyond improvements of a larger upper bound.	Jet generation through positivity and blow-ups; interpolation determinants and Taylor collisions.	Prescribing derivatives with weights fixed before the centres, while retaining determinant savings after clearing denominators.
Mahler inequalities (087)	Sharp lower volume products and equality classifications in all dimensions.	Gross’s conformal lens, complex mass estimates, cone–Laplace reduction, Gaussian covariance, and norm classification.	Exact boundary-feasibility normalization and zero-deficit rigidity; in the general route, simultaneous noncommuting normalization and global segment absorption.

Table 1 continued

Case	Barrier the conclusion would cross	Inherited or preceding machinery	Proposed step carrying the advance
Counting and sampling (113–115)	Unrestricted matching and common-base approximation; arbitrary-margin exact table generation.	Matching walks; restricted-variance simulation; quadratic signatures; dense completion and exact correction.	Full-gap amplification on the replicated matching space; restricted transport plus trace/restarts for common bases; table-law computation separating input size from precision.
Unique Games (102)	NP-hardness with near-one completeness and every fixed positive soundness target.	Shortcode inverse theory, Grassmann expansion, Fourier decoding, and conditional SDP consequences.	Nonlinear noise stability separated from linear detection on the output subspace, preserved through adaptive lifts and the decoder’s permitted information.
Direct finiteness (197)	A scalar characteristic-two one-sided inverse with nonzero reverse defect in an actual group algebra.	Finite-field interpolation, rank obstruction, character idempotents, and matrix-to-scalar transfer.	Joint realization of regular high-girth incidence, local subgroup embeddings, and packing that preserves the defect.
Complete Crouzeix bound (325)	Sharp constant 2 in every base and coefficient dimension.	Positive boundary calculus, Fourier/Gram estimates, and completely bounded similarity theory.	Ordered diagonal and cross estimates compatible under arbitrary amplification; in the structural route, sharp control of the optimal similarity condition number.
Relativistic Vlasov–Maxwell (362)	Large-data smooth existence without symmetry in the stated data class.	Retarded field representations, bounded-momentum continuation, and energy-doubling summation.	Signed source-force cancellation and bootstrap closure with scale-uniform constants and logarithmic momentum loss.

3 Review method and meanings of novelty

This is a purposive comparative narrative review. The catalogue supplied the family identifiers, while the principal manuscripts and their source files supplied theorem hypotheses, intermediate lemmas, and proof dependencies. Prior literature was identified from the manuscripts’ references and targeted searches for the closest earlier results. Comparison used primary papers, publisher records, author manuscripts, and archival preprints accessible on 7 October 2026. Bibliographic identities and supporting locations are recorded in the accompanying reference and source ledgers.

The sample contains ten family identifiers, grouped into the eight cases in Table 1: 007, 017, 087, 102, 113, 114, 115, 197, 325, and 362. Selection was contrastive and theme-guided. Family 087

supplies two distinct routes to sharp geometric inequalities; family 325 supplies direct and structural formulations; families 113–115 compare three different algorithmic transfers. Cancellation, positivity, and representation are organizing categories for this sample. The unit of analysis is a stated theorem and the proposed intermediate step carrying its additional scope. The accompanying comparison register records each inclusion, its primary baseline, and the boundary tested by the comparison.

Four questions guide each case. What is the mathematical obstruction? Which intermediate statement is intended to overcome it? Which tools were already available? Which additional scope, uniformity, or compatibility does the new argument require? This approach separates the *novelty of a conclusion* from the *novelty of its mechanism*. A familiar method can yield a major new theorem; an ingenious intermediate construction can remain valuable even if a subsequent implication needs revision. Historical comparison also separates a result’s generality from priority: manuscript dates and repository order alone do not establish precedence over concurrent work.

The comparison tracks what each construction must preserve: for example, an exact probability law, a nonzero algebraic defect, or an operator bound independent of coefficient dimension. It also tracks the order of quantifiers and the cost in dimension, precision, degree, or momentum. These details determine whether earlier results already achieve the same conclusion or whether a new estimate is needed.

The contrastive tests in Section 10.5 make three of these boundaries explicit. A restricted energy inequality can coexist with arbitrarily slow global mixing; an entrywise operator argument incurs a dimension factor; positive values on a finite mesh require additional control between its nodes. Together with the averaging and finite-field examples, these tests give precise content to the cross-field comparison.

Formal artifacts are considered through their documented mathematical statements and inspected source bindings. A comparator links a target statement to a solution module and permitted-axiom configuration [67]. The comparison asks whether its hypotheses and concrete objects match the manuscript theorem. Source ledgers record the statements, bindings, and execution status separately.

4 Arithmetic cancellation and Diophantine approximation

4.1 Ordinary multiplicative correlations: retaining the terminal scale

Family 007 addresses the difference between cancellation at a specified cutoff and cancellation after averaging over scales. Write $\lambda(n) = (-1)^{\Omega(n)}$ for the Liouville function, where $\Omega(n)$ counts prime factors with multiplicity. Its manuscript states that an absolute $c > 0$ gives, for each fixed pair of nonproportional affine forms $a_1n + b_1, a_2n + b_2$ with integer $a_i \geq 1$ and $b_i \geq 0$,

$$\left| \sum_{n \leq X} \lambda(a_1n + b_1) \lambda(a_2n + b_2) \right| \leq C_{a_1, a_2, b_1, b_2} X (\log X)^{-c}, \quad X \geq 3.$$

The coefficient-dependent constant can be ineffective; the statement fixes the forms before X grows [71]. The companion corrected Elliott statement concerns unit-disc-valued multiplicative functions. The obstruction is imitation of a periodic character multiplied by the oscillatory factor n^{it} ; the distance below measures this imitation on primes. One fixed factor must satisfy, for every fixed Dirichlet character χ , $\inf_{|t| \leq N} \mathbb{D}(f, \chi n^{it}; N) \rightarrow \infty$, where $\mathbb{D}^2(f, g; N) = \sum_{p \leq N} (1 - \Re(f(p)\overline{g(p)}))/p$. Under this hypothesis the ordinary two-point mean tends to zero, including fixed nonproportional affine forms. This second assertion supplies no rate uniform over that function class [71].

These are substantial changes of quantifiers relative to several established results. Matomäki and Radziwiłł control multiplicative means in most short intervals [46]; Matomäki, Radziwiłł

and Tao prove Chowla cancellation averaged over shifts [47]. Tao’s two-point theorem controls logarithmically weighted affine correlations [77]. Tao and Teräväinen obtain ordinary correlations outside an exceptional set of scales of logarithmic density zero [78]. Helfgott–Radziwiłł divisibility graphs and Pilatte’s subsequent prime-product construction deliver increasingly strong logarithmic cancellation [29, 74]. Pilatte already uses composite steps, centred prime factors and nonbacktracking walks. Consequently, the distinctive object of review is how the release combines these ingredients to reach every cutoff with controlled error.

Two closer comparisons sharpen the remaining barrier. Klurman, Mangerel and Teräväinen obtain ordinary binary cancellation on a set of full *upper* logarithmic density when one fixed factor is nonpretentious against every fixed character and fixed real twist [40, Definition 1.1 and Theorem 1.2]. Tao and Teräväinen also obtain a logarithmic-power saving for slowly growing affine Liouville forms outside a quantitatively controlled exceptional set of scales [79, Theorem 3.1 and Remarks 3.2]. The release’s quantitative assertion removes the exceptional cutoffs for fixed forms; the latter comparison retains greater coefficient uniformity. The release’s qualitative assertion combines every-cutoff cancellation with the stronger growing-twist hypothesis stated above. These are distinct changes of scope and hypothesis. The prospective breakthrough is completing the cutoff argument with its error budget intact, rather than obtaining an ordinary average or logarithmic saving in isolation.

One unusual bridge is a finite-law comparison through circuit complexity. The graph step uses products from disjoint prime supplies, with factors $\mathbf{1}_{p|n} - 1/p$. When the number of supplies grows like $\log \log X$, the joint residue pattern repeats only after the product of all participating primes, by the Chinese remainder theorem. That period is too large for direct uniform sampling. The manuscript instead encodes residue observables by binary variables and tests only bounded-depth circuits. Its finite-law lemma first corrects small low-order biases to obtain an *exactly* bounded-wise independent distribution, then applies Braverman’s circuit-fooling theorem. The correction has the established almost-independence precedent of Alon, Goldreich and Mansour [3, 11]. Crucially, translated sites retain the same residue for each prime. The comparison is applied to the signed trace expansion before absolute values; treating translated sites as independently resampled would change the arithmetic model. The manuscript’s finite-law, nonbacktracking-moment and mixed-difference lemmas give separately inspectable checkpoints [71].

Example 4.1 (Why translated sites share a prime residue). *Let U be uniform modulo a prime p , and set $c_0 = \mathbf{1}_{U=0} - 1/p$ and $c_h = \mathbf{1}_{U+h=0} - 1/p$. The same residue determines both translated sites, giving*

$$\mathbb{E}[c_0 c_h] = \begin{cases} (p-1)/p^2, & p \mid h, \\ -1/p^2, & p \nmid h. \end{cases}$$

Indeed, the two divisibility events coincide when $p \mid h$ and are disjoint otherwise. Independently resampling the residue at the second site would give zero in both cases, despite retaining each marginal law. This small calculation explains why the manuscript’s finite-law comparison keeps shared residues inside the signed trace: independence at the wrong level would erase the correlations being estimated.

The decisive quantitative bottleneck is uniformity of this comparison and the high-trace count as the number of prime supplies grows, including all squarefree padding choices. Repeated prime labels, rank witnesses and forest encodings must produce a saving with absolute constants rather than costs that erase the eventual logarithmic gain. A second, qualitative mechanism fixes its graph scale before taking the averaging cutoff to infinity. Its centre factors $\mathbf{1}_{p|n} - \theta/p$ need not have zero uniform mean. Vertex normalization and signed singleton integration provide the replacement

cancellation; first-visit tree bookkeeping retains return factors until averaging. This explains the order of limits and why the qualitative theorem need not assume a rate of nonpretentiousness [71].

The documented formal targets cover both the quantitative fixed-form assertion and the qualitative growing-twist assertion.¹

4.2 The exponent of π : logarithmic jets and determinant collisions

The irrationality exponent measures how well a real number can be approximated by rationals: larger exponents allow infinitely many unusually small approximation errors. Family 017 asserts the endpoint $\mu(\pi) = 2$. Its precise upper-bound statement is that, for every $\nu > 2$, some $Q(\nu)$ satisfies

$$|\pi - p/q| \geq q^{-\nu} \quad (p \in \mathbb{Z}, q \in \mathbb{N}, q \geq Q(\nu)).$$

The threshold is ineffective. The assertion concerns π , not π^2 , and does not imply a fixed lower bound c/q^2 or bounded partial quotients [72]. The contrast with established quantitative work is sharp: Zeilberger and Zudilin’s variant of Salikhov’s integral method proves the upper bound $7.103205334137\dots$ [83]. The released manuscript seeks the endpoint through a different geometric and arithmetic organization rather than optimizing those integrals.

Its central intermediate assertion is a separated-weight interpolation theorem. A jet is a finite collection of Taylor coefficients, so jet interpolation means prescribing values and derivatives at several points. The question is whether enough polynomials remain available after imposing the arithmetic constraints needed later in the proof. Polynomials use a full weighted simplex degree budget; after substituting $Y = 1 + t$ and $X_i = c_{ji} + u_i + \log(1 + t)$, they must prescribe specified jets at several coordinatewise distinct centres. The weights are selected successively, *before* the centres are known. The required degree threshold may subsequently depend on the centres. This asymmetry is essential: rational approximations determine centres later, so an argument choosing weights from those approximations would not supply the needed quantifier order [72].

Dimension counting alone does not establish this surjectivity. The manuscript proceeds through weighted transverse multiplicity, a curve-contact inequality, and positivity on an ordinary blow-up of finite-colength jet ideals. It then uses sufficiently large ordinary ideal powers and Serre vanishing to produce the prescribed jets. Roth-type zero estimates on products of commutative algebraic groups provide a relevant antecedent in Farhi [23]; Demailly relates curve positivity, blow-ups and asymptotic jet generation [17]. The release’s proposed bridge combines these themes for logarithmic curves with separated weights. The decisive geometric checkpoint is its curve inequality and the resulting centre-independent weight selection. The passage to ordinary ideal powers, rather than integral closures, is another exact point requiring review [72].

The arithmetic half constructs a nonzero rational-Gaussian interpolation minor whose denominator-clearing estimate supplies a lower bound. Translating its rows expresses entries through entire exponential-polynomial functions. Taylor degrees that coincide within a fixed transverse-index class create equal rows and annihilate that determinant term. Taylor-degree cancellation already underlies Laurent’s interpolation-determinant estimates [42]. Here the manuscript groups collisions by transverse indices and combines them with full-simplex counts: with m logarithmic coordinates, the growth exponents in the degree parameter are $m + 1$ for the polynomial-space dimension and m for the index-set cardinality. Many low indices yield collision savings; few yield powers of small

¹The comparator configuration selects three final declarations: `liouville_log_saving`, `binary_corrected_elliott` and `affine_corrected_elliott`. The inspected Lean wrappers bind them to the solution module `OAI.NumberTheory.TwoPointCorrelations.FinalMain`. Their assumptions match the stated fixed-form and nonpretentiousness scope. These declarations record the corresponding quantitative and qualitative conclusions.

approximation errors. The stated collision and determinant-comparison lemmas expose the analytic upper bound that must contradict the arithmetic lower bound [72].

A concrete mathematical consequence would be convergence of the Flint Hills series $\sum_{n \geq 1} (n^3 \sin^2 n)^{-1}$. Alekseyev proves that convergence requires $\mu(\pi) \leq 5/2$ [2]; Meiburg proves a sufficient strict inequality $\mu(\pi) < 5/2$ using sparsity of good rational approximations [48]. Thus the released exponent theorem would settle this convergence question without a borderline argument. The selected formal target records the approximation inequality and exponent identity.²

4.3 An elementary distinction between two averaging laws

Example 4.2 (A logarithmic average can miss a terminal-scale bias). *For $j \geq 1$, let $M_j = 2^{2^j}$, and put $a_n = 1$ on the disjoint integer intervals $[M_j, 2M_j]$ and $a_n = 0$ elsewhere. Each occupied interval contributes at most a fixed constant to $\sum_{n \leq X} a_n/n$, and there are $O(\log \log X)$ such intervals up to X . Consequently*

$$\frac{1}{\log X} \sum_{n \leq X} \frac{a_n}{n} \rightarrow 0.$$

At $X = 2M_j$, however, the ordinary mean is at least $1/2$. Thus logarithmic cancellation alone does not imply ordinary cancellation at every cutoff. This bounded-sequence illustration isolates the logical distinction; a multiplicative correlation requires its own arithmetic argument.

5 Convex duality and sharp operator bounds

The Mahler and Crouzeix families are useful tests of the release’s proposed mathematical mechanisms: both seek sharp universal inequalities, but the obstruction is not simply estimating another integral more accurately. For Mahler it is retaining enough information about polar geometry to identify the extremizers; for complete Crouzeix it is preserving noncommutative multiplication under arbitrary matrix amplification. The discussion below follows the released proof passages and identifies the transitions on which the claimed advances depend.

5.1 Mahler: boundary feasibility and Gaussian spectral comparison

A volume product measures how a convex body and its polar constrain one another; it is unchanged by invertible linear changes of coordinates. Sharp lower bounds therefore identify geometric shapes rather than a preferred coordinate system. For a compact convex body $K \subset \mathbb{R}^n$ with nonempty interior, write $K^\circ = \{y : \langle y, x \rangle \leq 1 \mid x \in K\}$ when the origin is interior. The symmetric manuscript claims, for every $n \geq 1$,

$$K = -K \implies |K| |K^\circ| \geq \frac{4^n}{n!},$$

with equality precisely for invertible linear images of Hanner bodies: bodies generated from centered intervals by Cartesian products and convex-hull joins. The general manuscript separately claims

$$\inf_{z \in \text{int } K} |K| |(K - z)^\circ| \geq \frac{(n+1)^{n+1}}{(n!)^2}, \quad \text{equality if and only if } K \text{ is a simplex.}$$

Neither statement assumes smoothness or strict convexity [60, 64]. The constants and extremal classes differ; the general inequality does not recover the sharp symmetric one by restriction.

²The comparator selects `OAI.PiExponent.main`. A separate, unselected declaration in `Main.lean` records Flint Hills summability.

The symmetric proof starts with a strip polytope $A = \{X : |b_i X| \leq 1\}$, using nonzero, pairwise nonproportional rows that span the dual space after removing redundant parallel strips. Its unusual connecting device is a conformal lens. A disk map F has boundary law

$$F(e^{i\Theta}) \stackrel{\text{law}}{=} \varepsilon \lambda(T) + iT, \quad T \sim \text{Unif}[-1, 1],$$

with an independent fair sign ε , and a strictly concave half-width λ . The manuscript explicitly obtains this map by rotating Gross's uniform-distribution example [26, Section 3.2]. Thus the conformal lens is a prior construction; the proposed advance is its use to convert complex feasibility probabilities into sharp real volume information.

For each independent row basis I , a boundary sample determines a point $Z_I = B_I^{-1}(F(e^{i\theta_i}))_{i \in I}$. Let P_I be the probability that every remaining strip coordinate belongs to the closed lens. The boundary-feasibility lemma claims $\sum_I P_I \geq 1$. Its proof applies a holomorphic isolated-zero mass estimate to the high powers $g(b_i z)^k$, with $g = F^{-1}$. The lower mass grows as k^n ; the substitution $u_i = \rho_i^{1/k} e^{i\theta_i}$ removes that factor and upper Fatou passes to closed boundary feasibility. This mass estimate belongs to the generalized Lelong-number framework [18, Sections 3 and 5]; complex-analytic approaches to reverse volume-product bounds also precede the release [8]. The noteworthy proposed step is the dimension-independent feasibility inequality with its exact normalization.

The next transition is more geometric. For fixed X , feasible signed bases give simplices in A° . Additional boundary ties are shown to have probability zero, and a supporting-functional argument proves that the simplex interiors are disjoint. The resulting identity is

$$\sum_I P_I = \frac{n!}{4^n} \int_A |\Sigma_X| dX \leq \frac{n!}{4^n} |A| |A^\circ|,$$

where Σ_X is their union. This exposes where the sharp constant enters. Approximation of arbitrary symmetric bodies by polars of inner polytopes then extends the bound. Compared with the established three-dimensional theorem and equality classification of Iriyeh–Shibata [31], the claimed scope is all dimensions through an analytic feasibility identity rather than a dimension-specific partition argument. All-dimensional nonsharp lower bounds also precede the release, including Kuperberg's indefinite-geometric and linking-integral approach [41]. The proposed qualitative gain is the conjectured sharp endpoint together with full rigidity, rather than a first all-dimensional reverse volume-product estimate.

Example 5.1 (A geometric transfer calibrated on the square). *Take $A = [-1, 1]^2$ with rows e_1, e_2 . There is one indexed basis, its feasibility probability is 1, and the four signed polar triangles fill the diamond A° , of area 2. Hence $(2!/4^2) \int_A |\Sigma_X| dX = (1/8) \cdot 4 \cdot 2 = 1$. Representing the same square by rows e_1, e_1, e_2 instead gives two indexed bases, $\{1, 3\}$ and $\{2, 3\}$, each with probability 1. Their geometric union remains the same diamond. Thus the probability sum is 2 while the normalized union volume is 1. Removing proportional rows is consequently part of the sharp geometric transfer: preserving the body alone does not preserve the indexed probability accounting. This checks the distinction between the feasibility inequality and the disjoint-simplex identity under their respective hypotheses.*

Equality requires a separate mechanism. After adjoining an ℓ_1 -segment, vanishing integrated uncovered volume is tested on product neighborhoods of a prescribed interior point and direction. Compactness gives a single feasibility witness, even when limiting row bases become dependent. The lens endpoint asymptotic $\lambda(1 - \eta) \sim (2/\pi)\eta \log(1/\eta)$ converts minimum representation costs into the existence of metric medians for every triple in the polar norm. The final three-ball classification

is the classical Hansen–Lima theorem [27, Corollary 7.4]. Hence the proposed rigidity advance lies in deriving that classical norm property from sharp volume equality. Checking the mass normalization, disjointness identity, and integral-to-universal-witness passage is more decisive than checking the concluding classification alone.

The general proof uses a substantially different structure. Lift the body to a closed, convex, solid, pointed cone $C \subset \mathbb{R}^m$, $m = n + 1$, and use its positive dual D . With interior vectors $U \in C, V \in D$ normalized by $\langle U, V \rangle = m$, define $\chi_C(V) = \int_C e^{-\langle V, x \rangle} dx$. The desired statement becomes $\chi_C(V)\chi_D(U) \geq 1$. This cone–Laplace correspondence is already explicit in Klartag’s work [39, Lemma 2.1 and Corollary 4.1]. The new claim concerns the estimate after that reduction, rather than the reduction itself.

Biased Gaussian projections onto the two cones supply cone-valued maps and projection derivatives $0 \preceq P_z \preceq I$. A simultaneous normalization chooses both linear coordinates and covariance feedback through a fixed-point argument, including a noncommuting matrix quadratic. Gaussian smoothing, injectivity, change of variables, and log-determinant concavity lower-bound the normalized logarithm of the Laplace product by $-\mathcal{E}$. Hermite-degree separation and inverse Ornstein–Uhlenbeck covariance representations then compare the entire layer field with spectral thresholds of a linear Gaussian matrix. The covariance machinery has primary precedents in Houdré–Privault [30, Proposition 2.1].

The purpose of $\mathcal{E}$ is to collect the remaining terms in this lower bound. Proving $\mathcal{E} \leq 0$ makes the logarithmic lower bound nonnegative and hence gives the desired product at least one. The hard step is therefore controlling these terms together, including those caused by noncommuting matrices.

A particularly distinctive ingredient is the tuned auxiliary curve: one component rotates with phase $4.6 \operatorname{arsinh}(x/3.6)$; another is chosen so the scalar profile plus the curve’s squared norm is affine in that component. A strict inequality for every nontrivial real segment absorbs divided-difference, layer, and commutator errors and establishes $\mathcal{E} \leq 0$. Equality forces commuting deterministic matrices, then diagonal cone-projection derivatives and an orthant cone, yielding a simplex section. This is global rigidity, beyond the prior local simplex minimality and stability theorem [38]. The delicate proposed advance is the simultaneous matrix normalization and all-segment absorption inequality, not Gaussian integration by parts in isolation. The June 2026 Chen–Li–Xi–Xu preprint already claims the general three-dimensional inequality and equality classification using shadow flows [12]; the release’s proposed extension is dimension-independent and follows a different proof architecture.

The scalar appendices provide exact rational node enclosures together with analytic interpolation and tail bounds. Their finite arithmetic certificates and the whole-real-line interpolation and tail arguments supply complementary checks: the former enclose specified nodes, while the latter connect those enclosures to the all-segment estimate used in the noncommuting matrix comparison.

5.2 Crouzeix: retaining order under matrix amplification

For $A \in M_n(\mathbb{C})$, let $W(A) = \{x^* A x : \|x\|_2 = 1\}$. The direct manuscript claims, for all positive n, m , all degrees $d \geq 0$, and arbitrary $B_k \in M_m(\mathbb{C})$,

$$\left\| \sum_{k=0}^d A^k \otimes B_k \right\|_2 \leq 2 \max_{z \in W(A)} \left\| \sum_{k=0}^d z^k B_k \right\|_2.$$

The coefficient dimension is independent of the base dimension. Scalar polynomials correspond to $m = 1$; allowing every m with the same constant is what makes the assertion *complete*. It controls

matrix-valued functions of A , whose coefficient products need not commute. Points and segments are included [59]. For a concrete sharpness control, take

$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad p(z) = z.$$

For a unit vector (x_1, x_2) , $x^*Ax = 2\overline{x_1}x_2$; its possible values fill the closed unit disk, while $\|A\|_2 = 2$. Thus constant two is already necessary for scalar coefficients. Sharpness and matrix amplification are separate parts of the proposed theorem. The established complete bound $1 + \sqrt{2}$ supplies the unconditional comparison [15]. Positive double-layer representations and convex spectral-set methods are older ingredients [6, 16].

The direct proof works first on an analytic convex enclosing domain Ω , with the matrix-valued analytic test normalized by $\sup_{z \in \Omega} \|F(z)\| \leq 1$. An exterior conformal coordinate produces Faber coefficients and a nonnegative boundary kernel having unit mass in both angular variables. The associated Fourier map is an L^2 contraction: the negative Fourier part is controlled by the positive part. This yields a Hilbert–Schmidt coefficient estimate with weight one for the constant coefficient and weight two for all positive degrees.

The decisive algebraic passage selects a top singular pair x, y of $F[A]$, compresses the positive Hermitian resolvent field, and tests the coefficient functional separately on FG and GF . These ordered tests bound both diagonal blocks using reciprocal Fourier weights. The off-diagonal block retains the constant term and has a matching lower bound. Positivity of the resulting 2-by-2 block forces $\|F[A]\| \leq 2$. The final block inequality is elementary; the substantive proposed improvement is obtaining its compatible diagonal and cross estimates uniformly in coefficient size.

This scope matters for current positioning. Lorist–Schwenninger’s August 2026 manuscript claims scalar constant two using uniformly bounded perturbations of 2-dilations and all powers f^j . Its Section 3(iv) explicitly identifies the commutativity obstruction to direct matrix amplification [43]. The September version of Åhag–Czyż–Virtanen claims the complete inequality for base matrices of order at most three [1]. Those preceding claims preclude describing every constant-two statement as new here; they do not by themselves supply the release’s arbitrary-dimensional complete theorem.

Two earlier scalar preprints are also close at the level of mechanism. Jin’s August version uses positive-real completion and ordered weighted Gramians; Luo’s August version uses top singular vectors, positive double layers, and Fourier or defect-Gram boundary certificates [34, 45]. Both claim scalar endpoints and expressly distinguish their scope from complete matrix-coefficient estimates. They narrow the novelty judgement: positive kernels, Gram certificates, and boundary cancellation already occur in preceding proposed routes. The release’s distinctive target is the compatibility of its ordered estimates under arbitrary matrix amplification.

The structural companion formulates the same sharp target through an attained minimum similarity: for a finite matrix whose numerical range lies inside a bounded convex domain with regular analytic Jordan boundary, the disk-coordinate operator is similar to a contraction with condition number at most two. A common continuous positive boundary density of mass I represents all independently chosen matrix-valued analytic tests [63]. Paulsen’s completely bounded similarity theorem already supplies the matching condition number and attained similarity for a unital completely bounded homomorphism [73]. The companion also derives its common density before establishing the sharp condition-number bound, using the classical contraction resolvent representation and crediting Delyon–Delyon. Similarity, attainment, and density existence are therefore inherited parts of this architecture. The candidate advance is the bound on the optimal condition number through the ordered density-comparison lemma; separate left and right multiplication estimates again preserve amplification. The companion also claims extension to bounded

operators on arbitrary complex Hilbert spaces and functions holomorphic near the closed numerical range, extending the target beyond finite-dimensional matrices.

Across both families, the proposed sharp constants depend on a compatibility that the classical ingredients do not separately supply: complex mass must match the disjoint polar simplices, scalar enclosures must control a global matrix comparison, and ordered Fourier estimates must survive amplification. This locates the mathematical contribution more precisely than attributing novelty to the use of positivity itself. The supplied formal targets describe the corresponding sharp statements.³

6 Counting, sampling, and the limits of local positivity

Families 113–115 form an integrated case study in converting structural inequalities into algorithms. Their common question is how to count or sample a constrained family when positive weights and feasible local moves alone give little quantitative information. Three different missing implications are addressed: an energy bound for a restricted class of functions must control the functions used by an algorithm; an approximate law must have a tractable exact description before it can be corrected; and a polynomial operation count must remain polynomial when integer data are written in binary. A fully polynomial randomized approximation scheme (FPRAS) estimates a count to relative error ε with high probability, in time polynomial in input size and $1/\varepsilon$. The question here is which new mathematical estimate would make such guarantees possible for the unrestricted families.

6.1 General perfect matchings: enlarge the space, then amplify the inequality

The older matching literature supplies both a method and a precise obstacle. Jerrum and Sinclair established approximation results through matching walks [32]; Jerrum, Sinclair and Vigoda obtained the permanent FPRAS by balancing hole-pattern weights [33]. The nonbipartite extension requires additional mathematics: Štefankovič, Vigoda and Wilmes show torpid mixing for the JSV-type chain under arbitrary hole-pattern weighting [80]. Their obstruction concerns that specified state space and transition mechanism. It is therefore compatible with an algorithm that changes both.

The release’s Theorem 1.1 covers finite simple undirected graphs, rational $0 < \varepsilon < 1$, and $0 < \delta < 1/2$, with relative-error failure at most δ , exact zero output on empty instances, and polynomial bit time on every random tape [52]. Its key sequence is Theorem 3.1’s hole-capacity bound, a weighted subdivision and colored tree-bag enlargement, and a sampler with local switches and exchanges between two matching coordinates. Theorem 6.1 bounds additive pair functions, rather than arbitrary pair functions. Lemma 7.2 supplies the missing upgrade: ordered product decomposition assigns each interaction residual to at most one coordinate pair, while replicated adjacent weight tiers make their accumulated energy absorbable. The replication count is $32L$, where L is the large polynomial comparison bound carried from the pair estimate. The manuscript derives its claimed polynomial-complexity bound using deliberately large parameters.

The distinctive review target is this change of analytic scale. A signed cycle identity is a comparison device, not a path of legal Markov moves; the full product-space inequality requires its own argument. Reading a two-coordinate estimate as full mixing would discard the step that makes the proposed construction unusual. The scope document lists the graph FPRAS, but only selected accompanying entropy and face statements [56]. Thus the algorithmic theorem and every geometric companion should retain their separate evidence status.

³The scope documents link comparator templates [61, 62]. In family 087, a sentence excluding nonsymmetric Mahler conflicts with the linked general-Mahler target; the target declares the general inequality and simplex equality. The comparison follows the target statement.

6.2 Common bases: restricted observables can be enough

A matroid abstracts the exchange properties of linear independence. Its bases are maximal independent sets; a common base must be a base of each of two matroids on the same ground set. An independence oracle answers whether a specified set is independent. For a single matroid, high-dimensional walks already yield an independence-oracle FPRAS [5]. Earlier common-base approximations included a deterministic $2^{O(r)}$ factor [4]. Their algebraic framework does not automatically survive intersection. On the four edges of $K_{2,2}$, the common bases of its two vertex-partition matroids have generating polynomial

$$g(x) = x_1x_4 + x_2x_3.$$

Its Hessian has eigenvalues $1, 1, -1, -1$; consequently it fails the one-positive-direction condition for a nonnegative quadratic Lorentzian polynomial. This elementary example isolates a structural obstruction, rather than a hardness conclusion.

The September manuscript’s Theorem 1.1 uses arbitrary equal-rank matroids on an explicitly enumerated ground set, with exact independence oracles. It bounds oracle calls and additional bit work polynomially on every execution [51]. A paired reduction uses $M_1 \oplus M_2^*$, then positive rank-deficiency weights and one-empty/one-full pair defects. Brändén–Huh quadratic signatures [10] control a recursively balanced signed current. Lemma 4.1 bounds transport between missing-slot conditional laws. Proposition 5.3 controls transversal values and defect-class means, leaving within-defect fluctuations uncontrolled. Lemma 6.2 separately bounds the transversal trace. Fresh warm restarts and observable variance estimates then support annealing and learned defect multipliers. The scope document specifies the independence-oracle matroid theorem [57]; it does not identify the later binary-capacity integer-polymatroid extension as part of that coverage.

Chen, Vigoda, and Yang already use partition-restricted Poincaré control to estimate hole-pattern frequencies for the permanent [14, Definition 4 and Theorem 5(ii)]. Their framework bounds stationary empirical averages of block-constant observables under the specified reversible-chain hypotheses. The common-base manuscript explicitly adopts this framework. Its candidate advance is the matroid-specific transport estimate together with compatible transversal trace and warm-start control. These additional estimates must connect the restricted inequality to the observables and returns used by the counting algorithm.

A contemporaneous comparison is also essential: Chen and Liu’s 5 October 2026 preprint also gives a common-base FPRAS and develops a nonnegative-curvature framework for Hadamard products of measures [13]. Its field-selection and curvature mechanism provides a separate comparison route. Manuscript dates alone therefore do not establish priority, and the September construction should be assessed through its specific transport and simulation arguments.

6.3 Contingency tables: make exactness affordable in expectation

The target law here is uniform over nonnegative integer tables with prescribed margins. Dense-margin sampling has longstanding geometric predecessors [20, 50], while perfect sampling was already available for two-row tables [37]. These restrictions matter: a result for fixed row count or sufficiently large margins does not give a polynomial bound jointly in both dimensions and the margin bit length.

The release’s Theorem 1.1 allows arbitrary nonnegative equal-total margins, both dimensions variable, and no cell restrictions. It supplies error 2^{-k} in worst-case polynomial bit time and exact uniform sampling in expected polynomial bit time [53]. Small-margin cells form explicit defect states; padding makes the remaining rectangular completion problems dense. Conditional transport

and repair lead to the small-state Poincaré inequality in Theorem 4.3. The exactness bottleneck is Proposition 6.3: the actual bounded sampler’s law, including fallbacks, is tabulated with an exponential input-size factor but polynomial dependence on precision. Theorem 6.4 applies the earlier residual-mixture construction of Göbel, Liu, Manurangsi and Pappik [25]. Their method requires usable output-law computation, not merely a small total-variation error.

For any proposed exact correction, write δ for the actual correction-branch probability and T_{ordinary} for the mean work conditioned on the ordinary branch. The decisive accounting is

$$\mathbb{E}[T] = (1 - \delta)T_{\text{ordinary}} + \delta \mathbb{E}[T_{\text{correction}}].$$

The correction branch may be expensive while its contribution to the expectation remains controlled. An implementation that instead enumerates all precision-dependent random tapes can destroy that accounting. This distinguishes a probability identity from an efficient sampler. The scope document likewise separates unrestricted exact sampling from cell-bounded approximate counting [58]; forbidden cells cannot simply be inserted into the unrestricted sampler’s guarantee.

6.4 From algorithmic guarantees to implementations

The three constructions repair different interfaces between analysis and computation. Matching replication upgrades a restricted inequality; common-base trace simulation exploits a restricted inequality without that upgrade; table-law tabulation makes a rare exact correction computable. Calling all three *rapid mixing from positivity* would obscure the specific work.

A useful research program would isolate these interfaces as reusable lemmas with independently checked hypotheses. For common bases, one could compare conditional transport and curvature on the same oracle families, measuring which quantities each certificate controls. For tables, the natural implementation task is a finite-kernel law evaluator whose state enumeration is separated from its requested precision. For matchings, a small-instance prototype should report enlarged graph size, replication, and acceptance probabilities before extrapolating runtime. These are proposals motivated by the proof structures, not performance conclusions.

The theorem statements concern explicitly enumerated graphs or ground sets, exact oracles, and binary integer input. They provide an algorithmic frontier if the proposed proofs hold. Practical utility additionally requires feasible constants, a faithful target distribution, and measured cost against existing restricted-family algorithms. The scope documents identify the supplied formalization targets; assessing the supplied proof architecture and running a proof checker are distinct verification activities.

7 Unique Games and the completeness barrier

A Unique Game assigns labels to vertices, with each edge requiring one label to be a specified bijective function of the other. The goal is to satisfy as many edges as possible. Completeness describes the fraction achievable in the yes instances of a reduction; soundness bounds the fraction achievable in its no instances. Family 102 claims hardness even when these fractions are near one and near zero, respectively. The established shortcode and Grassmann-expansion line gives imperfect-completeness hardness [7, 36]. Near-one completeness is a separate requirement. Ordinary repetition multiplies completeness as well as reducing soundness, so it does not itself repair a one-half completeness barrier. Khot’s survey states the independent-error quantifiers of Unique Games [35], and Raghavendra’s SDP-optimality framework gives consequences conditional on that hypothesis [75].

The release’s Theorem 1.1 asserts a deterministic polynomial-time reduction from 3SAT, for every fixed $0 < \varepsilon, \delta < 1/2$, to simple unweighted bipartite translation games over $\mathbb{F}_2^s$, where $s = s(\varepsilon, \delta)$ depends only on the fixed error parameters, with completeness at least $1 - \varepsilon$ and soundness at most δ [54]. Lemma 3.1 proposes a latent-alphabet gadget: an equivariant nonlinear observable is stable under noise, while families of linear observations with sufficiently high restriction rank on the gadget’s distinguished output subspace K detect that noise with probability at least $1/8$. Section 5 uses Fourier comparison to return to the ordinary degree-two shortcode inverse theorem. A correlated-advice decoder and clean-coordinate repetition supply the soundness bridge. The scope document describes the translation-game theorem and separately lists Max-Cut, cover and deletion reductions [55].

The gadget seeks to keep a nonlinear signal stable while making a sufficiently rich family of linear probes detect the perturbation. This is the proposed way to preserve high completeness without losing the soundness argument. The new object is that simultaneous separation, while the consequences for SDP thresholds have an established conditional theory. Its evaluation should concentrate on uniformity over adaptive linear lifts, the row-fiber transfer, and the information actually visible to each decoded strategy. Håstad’s earlier Fourier decoding explains why extracting a strategy need not mean recovering a nearby codeword [28]; the new gadget must still preserve the particular completeness and soundness required here.

The distinction from a routine amplification step is thus mathematical: ordinary repetition reduces both completeness and soundness, whereas this gadget is intended to separate their behavior. That gain depends on stability under adaptive linear lifts and on a decoder that uses only the information available to a valid strategy.

8 Finite incidence data and one-sided inverses in group algebras

A unital ring is directly finite when $ab = 1$ implies $ba = 1$. Finite-dimensional algebras satisfy this property by linear algebra. For group algebras in characteristic zero, finite tracial operator methods replace finite dimension [49]. In positive characteristic, Elek and Szabó established direct finiteness for sofic groups using finite approximation and rank methods [22]. Thus a genuine counterexample over a field of positive characteristic also tests the reach of finite approximate models of groups.

The characteristic-two manuscript in family 197 states a scalar counterexample over a finite field K , with a finitely presented group containing odd-order torsion [65]. Its construction uses a finite incidence system, local subgroup embeddings, and character packing. The inputs include vectors $x_v \in K^m$ with

$$\sum_{v \in f} x_v x_v^\top = 0 \quad (f \in \mathcal{F}), \quad \sum_{v \in V} x_v x_v^\top = -I_m, \quad 0 < m < |V| \leq m\ell^2. \quad (1)$$

The regular incidence degree is $\ell + 1$, with ℓ an odd prime and K containing a primitive ℓ th root of unity, and the incidence graph has girth at least 12. Low-degree finite-field interpolation supplies the pairing identities; a punctured affine-line construction supplies the geometric compatibility. The girth condition preserves embedded local groups. Character idempotents then pack a rectangular inverse relation into square matrices, whose reverse defect is detected inside an embedded subgroup algebra. A finite auxiliary group converts the matrix obstruction into scalar elements. The difficult part is making these stages compatible: a local relation must survive embedding and still leave a detectable reverse defect.

Two elementary observations help locate its mathematical substance. First, finite support does not mean finite dimension: multiplication can move outside the span of the original supports.

Second, local cancellation can coexist with a nonzero total sum in characteristic two. If every point occurs in $D = \ell + 1$ members, summing all local identities in (1) gives

$$0 = D \sum_{v \in V} x_v x_v^\top,$$

and $D = 0$ as a field scalar. There is therefore no contradiction with a nonzero total pairing. The construction makes essential use of the coefficient characteristic, rather than importing a real positive semidefinite interpretation of $x_v x_v^\top$.

Example 8.1 (A miniature finite-field cancellation). Write $\mathbb{F}_4 = \{0, 1, \alpha, 1 + \alpha\}$ with $\alpha^2 + \alpha + 1 = 0$. The degree-one interpolant $f(t) = 1 + t$ satisfies

$$\sum_{t \in \mathbb{F}_4} f(t)^2 = 0, \quad \sum_{t \in \mathbb{F}_4 \setminus \{0\}} f(t)^2 = 1.$$

Deleting the point at which $f(0) = 1$ changes a vanishing full-field sum into a nonzero pairing. This illustrates the algebra behind puncturing. The high-girth regular incidence system and subgroup embeddings are the additional ingredients needed for the manuscript's counterexample.

Stable finiteness concerns every matrix ring and is stronger than direct finiteness of a fixed ring. Dykema and Juschenko proved that stable finiteness of $K[G]$ is equivalent to direct finiteness of $K[G \times H]$ for every finite group H [21]. The transfer from matrices to scalars therefore has an established conceptual predecessor. The candidate new construction is the incidence system that supplies the defect while allowing all required subgroup embeddings and character packing. Its regularity and high girth serve different purposes: the first makes the characteristic-two cancellation possible, while the second protects local relations from unintended identifications. The documented selected formal scope covers the characteristic-two scalar defect; its nonsofic consequence follows mathematically from Elek and Szabó [22] once that defect is established [69]. The original characteristic-two manuscript and the catalogue's later torsion-free companion address different group hypotheses and should be distinguished.

9 Signed impulse and large-data relativistic Vlasov–Maxwell

For particle momentum v , put $q(v) = \sqrt{1 + |v|^2}$ and $u(v) = v/q(v)$. The transport equation is

$$\partial_t f + u(v) \cdot \nabla_x f + (E + u(v) \times B) \cdot \nabla_v f = 0,$$

coupled to Maxwell's equations with charge and current generated by f . Global weak existence and global classical regularity are different problems. The former has a longstanding theory [19, 76]; the latter requires control that prevents concentration or unbounded momentum support.

Glassey and Strauss established a bounded-momentum continuation criterion [24]. Bouchut, Golse, and Pallard developed a physical-space proof [9], and Luk and Strain weakened the required support control to its projection onto a fixed two-dimensional plane [44]. Wang's July 2026 two-paper treatment addresses arbitrarily large cylindrically symmetric localized data [81, 82]. This recent comparison matters: the proposed additional generality is removal of symmetry within the stated data class, rather than large-data existence as an undifferentiated label.

Family 362 states global smooth classical existence and uniqueness for one-species three-dimensional data with nonnegative compactly supported smooth particle density, finite-energy

smooth initial fields with bounded derivatives, and the Gauss constraints [66]. Its key quantity is the *signed impulse* along a particle characteristic,

$$V(t_2) - V(t_1) = \int_{t_1}^{t_2} (E(t, X(t)) + u(V(t)) \times B(t, X(t))) dt.$$

An asserted source-trajectory cancellation rewrites a singular retarded force contribution as a total derivative and residual terms. A direction estimate improves occupation control before the signed-force selection uses it. Fix a finite horizon T_0 and a compact classical solution prefix. The parameters P and w are dyadic, with P above a fixed threshold and at least 1024 times the maximum supported particle energy on that prefix. On an interval throughout which $w/8 \leq q(V_X(t)) \leq 8w$, the resulting bootstrap aims to prove

$$|V(t_2) - V(t_1)| \leq MP\sqrt{\Delta t} + A\Delta t \frac{P^2 \log(2 + P)}{w}, \quad (2)$$

The constants M, A may depend on the datum and T_0 , but are uniform in the compact solution prefix and the parameter P . The decisive new burden is closing this estimate, including cutoff and receiver-acceleration errors.

The terminal implication can be inspected separately. Suppose a finite maximal lifespan exists, fix that lifespan as T_0 , and let t_n be the first time the maximum supported energy reaches 2^n . Choose a label attaining this maximum and let s_n be its last crossing of 2^{n-1} before t_n . Then $s_n \geq t_{n-1}$, and its energy stays in $[2^{n-1}, 2^n]$ on $[s_n, t_n]$. Take $P = 1024 \cdot 2^n$, $w = 2^n$, and $\Delta t = t_n - s_n$; all band and prefix hypotheses hold for sufficiently large n . Since q is 1-Lipschitz, that energy increase requires a momentum increment at least 2^{n-1} . Substituting in (2) and cancelling 2^n gives

$$\frac{1}{2} \leq 1024M\sqrt{\Delta t} + 1024^2 A\Delta t \log(2 + 1024 \cdot 2^n).$$

At least one term is at least $1/4$, so $t_n - t_{n-1} \geq \Delta t \geq c/(n+1)$. The sum diverges. Conditional on the uniform impulse estimate and continuation criterion, infinitely many doublings cannot occur in finite time. This isolates the analytic innovation from its final continuation consequence.

The momentum loss enters this route through a summability test. Replacing $\log(2 + P)$ in (2) by an increasing $H(P) \geq 1$ would give $t_n - t_{n-1} \geq c/H(1024 \cdot 2^n)$. Divergence of the reciprocal-scale series therefore suffices for the same continuation inference. In particular, the loss $\log^\beta(2 + P)$ with $\beta > 1$ would supply only summable lower bounds of order $(n+1)^{-\beta}$. A mildly larger loss of order $\log P \log \log P$, by contrast, still gives the divergent series $\sum_n 1/(n \log n)$. The proposed methodological advance is the signed-force closure with a divergence-compatible budget and scale-uniform constants; the final summability argument is its elementary consequence.

Taking absolute values before integration would replace the left side by an accumulated force norm and discard the cancellation. The general lesson is shared with correlation estimates in number theory, although the actual identities and geometric estimates are different. Here the distinction from a standard refinement is clear: a smaller constant in an absolute-force bound would retain the same obstacle, whereas a signed bound with a divergence-compatible momentum loss would change the continuation argument. Its validity depends on the source-time change of variables, angular sums, and bootstrap constants remaining uniform as momentum grows. The selected formal statement includes the smooth global solution and uniqueness class [70].

10 Methodological novelty across the selected cases

The cases differ sharply in what would be new. In the arithmetic and kinetic problems, the central proposal is stronger cancellation with the right uniformity. In the operator problem, it is

compatibility of ordered estimates under arbitrary amplification. In the algebraic and algorithmic cases, it is a construction that preserves a defect or a probability law through several changes of representation. These are more informative descriptions than the names of the conjectures alone.

10.1 Cancellation before irreversible estimates

The arithmetic and kinetic cases share an order-of-operations principle. An absolute-value estimate is irreversible: a small signed contribution can become a large positive contribution once its sign is discarded. In the correlation manuscripts, common prime residues and repeated labels retain dependencies that contribute to cancellation. In the kinetic manuscript, source-trajectory differentiation combines transport and acceleration terms before the force norm is estimated. Their detailed mechanisms differ, but in each case the proposed advance depends on keeping the relevant structure until the averaging or integration step.

This observation suggests a disciplined search for transferable ideas. The candidate is a representation that makes a signed identity visible, together with a norm estimate for its residual terms. A cancellation identity alone does not control the boundary, cutoff, or dependence costs created by using it. The most useful research question is consequently whether the resulting residual budget closes with the quantifiers required by the application. The high-trace and angular-summation checkpoints offer concrete examples of this distinction.

10.2 Positivity has several mathematically distinct meanings

Positivity of a scalar expression, positivity of an operator block, and a one-positive-direction signature are different statements. The Crouzeix case uses operator positivity in a functional calculus that must remain valid under amplification. The matroid case uses an indefinite quadratic structure to transport selected observables. The convex-geometric cases convert projection or feasibility information into an integral inequality. These uses should be compared at the level of their exact hypotheses, rather than grouped under a universal positivity principle.

The finite-field example offers a useful contrast. Transpose pairings in characteristic two have no real order structure: simultaneous vanishing local sums and a nonzero total sum are algebraically compatible. Treating the pairings as positive semidefinite matrices over the reals would destroy the intended construction. The review's connection is methodological: each argument chooses a structure whose composition rules can carry the desired global conclusion.

10.3 Changing representation changes the proof obligation

The counting case studies do more than optimize a familiar chain. They alter the state space, the observable family, or the precision model in which a transfer theorem is applied. The group-algebra construction moves from finite incidence to local groups, rectangular relations, square matrices, and finally scalar group-ring elements. The operator argument must retain the sharp constant under passage to matrix-valued functional calculus. Each movement creates an interface at which a property must be preserved.

For sampling, that property is a probability law: copying labels can change how often an unlabelled object occurs. For algebra, it is a nonzero defect: a map available on a subgroup need not extend to the whole ring. For complete operator bounds, it is a dimension-independent norm statement: entrywise scalar estimates can introduce a coefficient dimension factor. A proposed reduction therefore deserves attention both for what it simplifies and for what it must preserve.

10.4 Uniformity can be the main mathematical advance

Several claims hinge on constants, quantifiers, or limits that are easy to lose in informal exposition. Ordinary cancellation at every cutoff is stronger than cancellation outside an exceptional set of scales. Selecting interpolation weights before the approximation centres are chosen is stronger than choosing a convenient weight after seeing those centres. A momentum estimate with constants independent of the current solution interval can rule out finite-time breakdown; a bound depending on that interval's uncontrolled force cannot supply the same continuation argument.

Computational uniformity is equally substantive. A polynomial number of oracle calls, a polynomial bit bound, expected work, and a bounded-time algorithm describe different guarantees. An exact output distribution is a stronger law statement than a total-variation approximation, but its expected cost can include exceptionally expensive correction events. The question for a practical application is which of these quantities remains controlled on the intended instances.

10.5 Elementary tests of the mathematical distinctions

The following elementary examples make the comparison concrete. Each isolates a boundary between a useful intermediate statement and a stronger conclusion required by one of the reviewed arguments.

Example 10.1 (Restricted energy control and the full spectral gap). *On $\{-1, 1\}^2$ with uniform law, consider the continuous-time chain that flips the first coordinate at rate 1 and the second at rate $\tau > 0$. Its Dirichlet form is*

$$\mathcal{E}_\tau(f) = \frac{1}{2} \mathbb{E} \left[(f(-a, b) - f(a, b))^2 + \tau (f(a, -b) - f(a, b))^2 \right].$$

For every observable $f = g(a)$, one has $\mathcal{E}_\tau(f) = 2 \text{Var}(f)$, uniformly in τ . For $f(a, b) = b$, however, the variance is 1 and the energy is 2τ . The functions 1, a, b, ab diagonalize the negative generator, with eigenvalues 0, 2, $2\tau, 2(1 + \tau)$. Thus the full spectral gap is $2 \min\{1, \tau\}$, although the restricted inequality remains fixed. At $\tau = 0$ the restricted inequality even coexists with reducibility. This locates the exact distinction between the matching argument's upgrade to a full inequality and the common-base argument's separate use of selected observables and return control. A restricted estimate must be paired with the additional property the algorithm actually uses.

Example 10.2 (The dimension loss in an entrywise operator estimate). *Suppose the scalar inequality $\|p(A)\| \leq C \max_{z \in W(A)} |p(z)|$ is available, and let $Q(z) = [q_{ij}(z)]_{i,j=1}^m$ be a matrix polynomial. Viewing $Q[A]$ as a block matrix gives the elementary estimate*

$$\|Q[A]\| \leq m \max_{i,j} \|q_{ij}(A)\| \leq mC \max_{z \in W(A)} \|Q(z)\|.$$

Indeed, bound each output block by the largest block norm times $\sum_j \|x_j\|$, then apply Cauchy–Schwarz to input and output blocks. This argument retains a factor m . The complete Crouzeix claim seeks constant 2 independently of m ; its ordered product tests must remove the dimensional loss that this entrywise route leaves behind. The distinction concerns the strength of the argument, even when its scalar inputs are already sharp.

Example 10.3 (Positive nodes and a continuum inequality). *For $h > 0$, the polynomial $r_h(t) = (t - h/2)^2 - h^2/16$ is positive at every node $t \in h\mathbb{Z}$, where $r_h(t) \geq 3h^2/16$, but $r_h(h/2) = -h^2/16$. A positive node certificate therefore needs a separate argument between nodes. For example, on a covered compact interval, a derivative bound $|r'| \leq L$ and node lower bounds $r \geq Lh/2$ imply $r \geq 0$ whenever every point is within $h/2$ of a node. An all-real conclusion additionally needs control*

beyond the covered interval. This explains the distinct roles of rational node enclosures, interpolation, and tail estimates in the general Mahler manuscript. The proposed advance is their compatibility with the scalar error budget needed by the matrix comparison.

11 Research consequences and application pathways

The most promising further work begins with a specific mathematical object. In correlations it is the finite-law comparison that preserves shared prime residues; in kinetic theory it is the source-force identity and the size of its residual terms. The similarity and exact-correction arguments, by contrast, already have general theories. Their value here depends on new estimates or computable laws that meet those theories' hypotheses. This distinction helps identify which parts merit extraction as reusable lemmas.

Three application routes have especially concrete tests. A matching prototype can measure how weighted subdivision and replication enlarge the input before its sampler is run. A table sampler can compare the uniform output law and expected correction cost against methods for restricted margins. An operator implementation can compare the sharp constant with the established $1 + \sqrt{2}$ bound on the same numerical-range enclosure. In each case, the theoretical improvement matters only to the extent that its cost or error remains significant after the surrounding computation. The PDE claim would primarily change the analytical foundation; a numerical method would still need its own stability and convergence analysis.

The finite examples suggest manageable first questions without replacing the main proofs. The Mahler scalar certificates can be examined together with the interpolation and tail estimates that extend them beyond their nodes. Common-base transport and curvature can be compared on the same oracle families. A finite incidence witness can be checked against the regularity, pairing, and embedding conditions simultaneously. Such studies would explain which parts of the proposed arguments travel to other problems and which rely on their original setting.

12 Conclusion

A large claimed theorem and a new mathematical method are different achievements. The selected release manuscripts would have major consequences if established, but much of their surrounding machinery is classical. Their candidate methodological contributions are more specific: retaining shared-residue cancellation at every cutoff, choosing interpolation weights before the centres, controlling ordered operator estimates in every coefficient dimension, constructing compatible incidence data, and closing a signed-force estimate across all momentum scales.

The common lesson is that a seemingly small condition can mark a qualitative boundary. A restricted energy estimate is useful only for the observables an algorithm needs; a finite certificate reaches a continuum only through additional control; retaining a sharp scalar constant for matrix coefficients requires additional control under amplification. The review locates these boundaries and compares the proposed ways of crossing them. They offer a more useful guide to the release's interesting mathematics than treating every sophisticated ingredient as an innovation.

AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

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